

LECTURE / SESSION¹ PLAN

Name Lecturer: Sofia Gil-Clavel	Date session: 3	Expected number of students: 20
Course title	R tools for teaching and research	
Topic of lecture or session	VU tools for teaching and research: JupyterHub and AmCAT.	
Situational factors (e.g. group size, prior knowledge, expected motivation)	The class lasts 3hrs. The students have at least basic knowledge of R and tidyverse. The students use different Operating Systems (Windows, Mac, or Linux).	
Intended learning outcomes of this session	At the end of the session the students will be able to: 1. Explain what a server is. 2. Explain what JupyterHub is. 3. Upload and download files to JupyterHub. 4. Explain what the Amsterdam Content Analysis (AmCAT) is. 5. Use AmCAT to share, access, and analyze data.	
Learning material (book, chapters, ...)	The lecture is based on these sources: • Gil-Clavel, Sofia. <i>Constrained by Computing Power? We Can Help You!</i> January 2025. https://societal-analytics.nl/blogs/20250201_computing-power/. • Gil-Clavel, Sofia. "AmCAT Workshop - COMPTXT 2026." <i>Societal Analytics Lab - Blogs</i>, April 2026. https://societal-analytics.nl/blogs/20260422_AmCAT_Workshop/.	
Media, equipment, tools	The students use their own laptops. The teacher needs access to a projector and a whiteboard.	

¹ A session means a teaching and learning session for university students in the bachelor or master. It can be a lecture, a seminar or a specific type of educational meeting with a group of students that is relevant for the discipline where the lecturer can demonstrate a whole range of teaching skills. It means that in one session each of the six didactic elements appears at least once. Your lesson plan should show that you are using a powerful learning environment and that students are activated.

Preparation for students	The students have access to the slides and codes before the class: https://github.com/SofiaG11/Course_Rtools/3_VUtools
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Time ² (min.)	Didactic element (goal) ³ and topic ⁴	What the teacher does ⁵ (teacher activity)	What students do (learner activity)	Evaluation ⁶ (feedback/assessment)
20	Introduction to servers.	The teacher uses the slides to explain what a server is.	The students passively digest what the teacher is explaining.	The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.
20	Understand and explain what JupyterHub is and how to use it to standardize students' R-environments.	- The teacher uses the slides to explain what JupyterHub is and how to access it. - The teacher explains the process to request a JupyterHub environment for their lectures.	The students passively digest what the teacher is explaining.	The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.

² Indicate the planned duration in minutes

³ State the number(s) of the relevant ILO at each didactic element

⁴ Only key words

⁵ Specify the type of activity and write down also the questions that you prepared to ask in the session

⁶ Specify the type of evaluation, i.e. the way in which you assess if the objective(s) has/have been achieved

40	How to upload and download data to/from VU JupyterHub.	- The teacher uses the slides to explain how to access JupyterHub for Windows users. - The teacher uses the slides to explain how to access JupyterHub for MAC users.	The students follow the steps in their computers.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
40	Explain what the Amsterdam Content Analysis (AmCAT) is.	- The teacher explains what AmCAT is. - The teacher explains how AmCAT facilitates sharing, searching, and analyzing data. - The teacher explains how to: - access AmCAT - request a project - upload data - share data while keeping full control	- The students passively digest what the teacher is explaining. - The students follow the steps in their computers.	- The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student. - The teacher walks around the classroom to check on the students and provides feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
15	Explain AmCAT connection to VU JupyterHub.	The teacher explains how AmCAT can be used in the context of teaching.	The students passively digest what the teacher is explaining.	The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these

				confusions the teacher will be able to correct the student.
40	• Explain how to access AmCAT using the API.	• The teacher explains how to use the AmCAT APIs in R to: ○ access data ○ analyze data	• The students follow the steps in their computers.	• The teacher walks around the classroom to check on the students and provides feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.